GFN

Microsoft Azure

5

03.07.2026

Tag 10

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

GitHub Repo https://github.com/www42/a5

AZ-104

Learning Paths:

- ➡️ Manage identities and governance in Azure
- ➡️ Prerequisites for Azure administrators – CloudShell and templates
- ➡️ Configure and manage virtual networks for Azure administrators
- ➡️ Implement and manage storage in Azure
- ➡️ Deploy and manage Azure compute resources
- ➡️ Monitor and back up Azure resources

What happened yesterday

Maaßstr. 24
69123 Heidelberg

Maaßstr. 24
69123 Heidelberg

Maaßstr. 24
69123 Heidelberg

Entymologie ist sehr wichtig, finde ich. Nenne 5 Beispiele aus der IT, die griechischen Ursprungs sind. Namen, Begriffe, Protokolle. Z.B. Kerberos.

Recherchierte fünf IT-Begriffe mit griechischen Wurzeln

Gerne! Etymologie ist wirklich spannend, gerade in der IT gibt's viele griechische Wurzeln. Hier fünf Beispiele:

1. **Kerberos** - benannt nach dem dreiköpfigen Höllenhund (Κέρβερος) aus der griechischen Mythologie, der den Eingang zur Unterwelt bewacht. Passend fürs Authentifizierungsprotokoll, das drei Parteien (Client, Server, KDC) absichert.
2. **Hypertext** / **HTTP** - "hyper" von griechisch ὑπέρ (hypér) = "über, hinaus". Bezeichnet Text, der über die lineare Struktur hinausgeht, also verlinkt ist.
3. **Cryptography (Kryptographie)** - von κρυπτός (kryptós) = "verborgen" und γράφειν (gráphein) = "schreiben". Also wörtlich "verborgenes Schreiben".
4. **Daemon** - kommt von δαίμων (daímōn), ursprünglich ein neutraler Geist oder eine Zwischeninstanz zwischen Göttern und Menschen (nicht zwingend böse, wie später im Christentum umgedeutet). In Unix bezeichnet ein Daemon einen Hintergrundprozess, der quasi "im Hintergrund wirkt", ohne dass man ihn direkt sieht.
5. **Cybernetics** / **Cyberspace** - von κυβερνήτης (kybernétēs) = "Steuermann". Norbert Wieners "Kybernetik" (1948) leitete daraus den Begriff für Steuerungs- und Regelsysteme ab, und daraus entstand später "Cyberspace".

Als kleinen Bonus, falls es dich interessiert: **Ethernet** trägt "Ether" (αἰθήρ, aithḗr) im Namen - den "Äther", das antike Konzept eines Mediums, das den ganzen Raum durchdringt. Passend für ein Netzwerkprotokoll, das (theoretisch) alle Geräte im lokalen Netz "durchdringt".

Write a message...

Sonnet 5 Medium

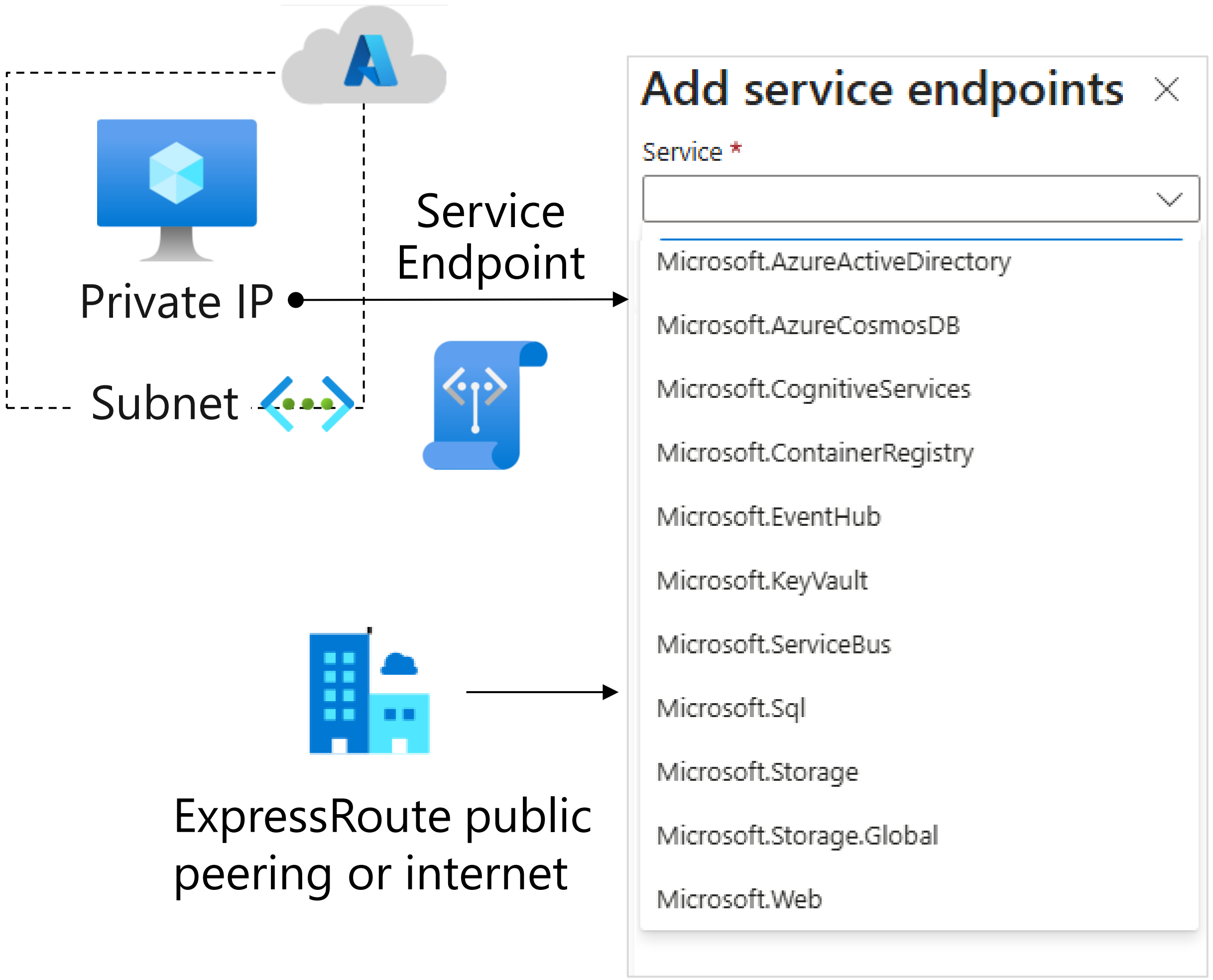
AZ-104

➡️ Configure and manage virtual networks for Azure administrators

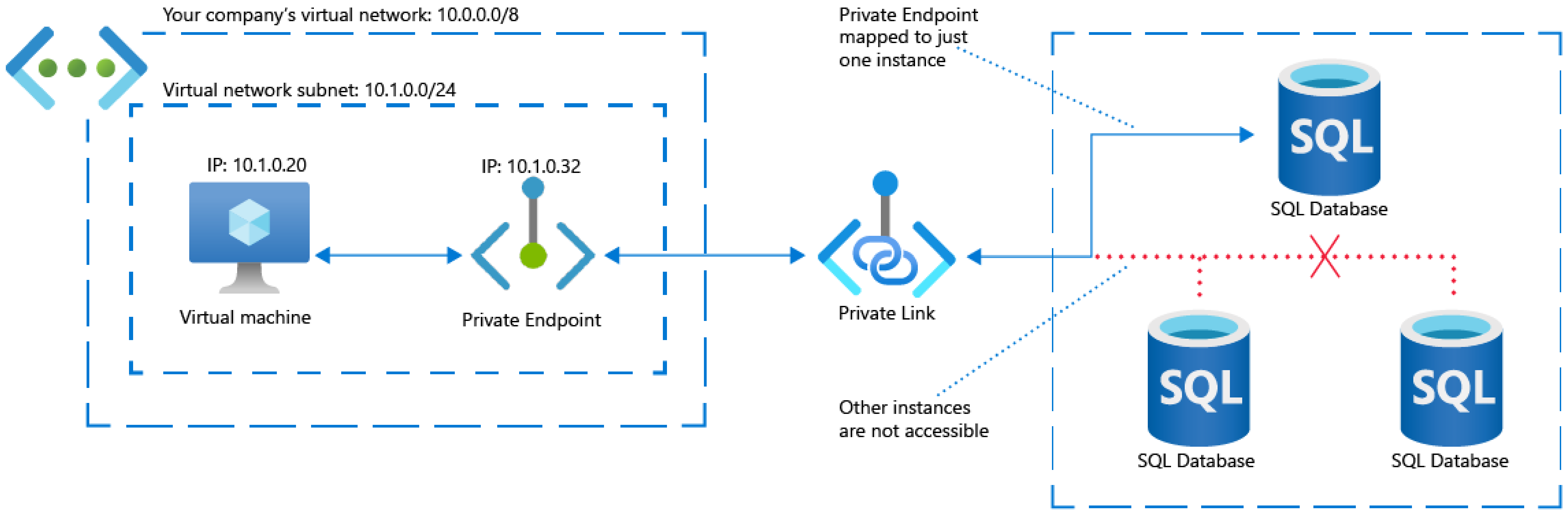
Service Endpoints and Private Endpoints

What is a Service Endpoint?

- A direct connection from your virtual network to supported Azure services
- Restricts the flow of traffic and enables your Azure VMs to access the service directly from your private address space
- Traffic always stays on the Microsoft Azure backbone network



What is an Azure Private Endpoint?



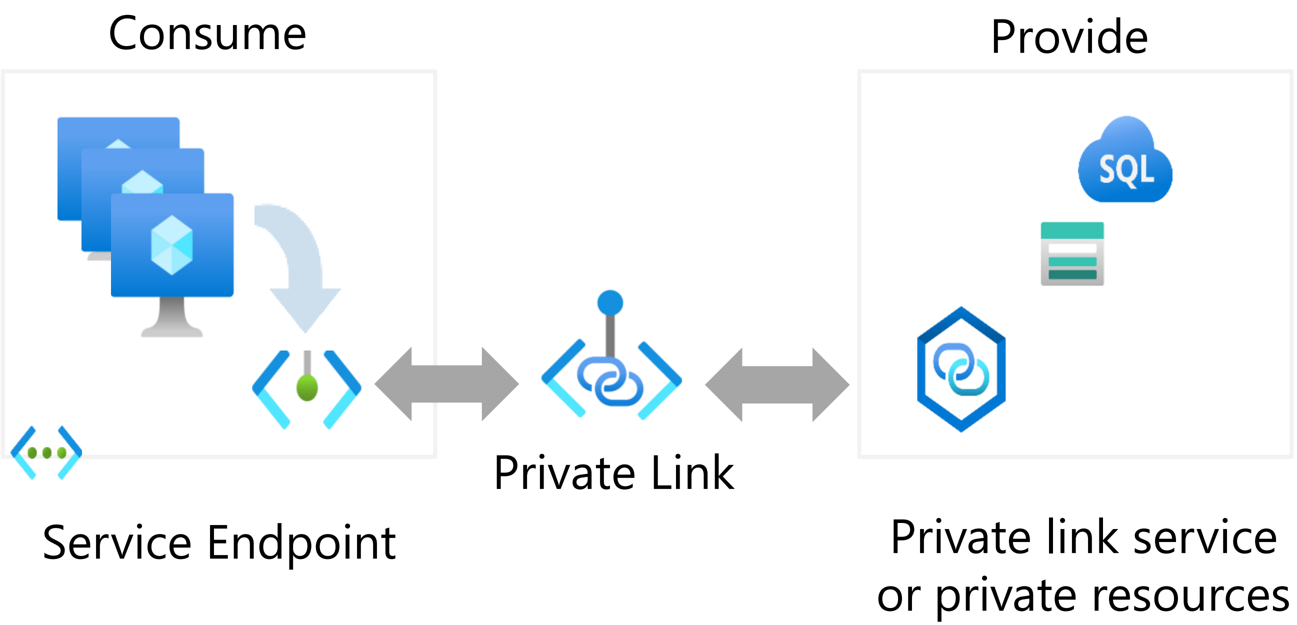
Uses a private IP address from your virtual network

Granular access to services

Eliminates the potential security risk of a public IP address

What is Azure Private Link?

- Build and allow private connections between resources
- To privately connect to a service, create a service endpoint
- To privately render a service, create a private link service or private resources
- All data travels on the Microsoft backend



Compare Service Endpoints and Private Endpoints

Feature	Service Endpoint	Private Endpoint
Use Case	Simple, cost-effective way to secure traffic to Azure services from within VNet	Maximum security, regulatory compliance, hybrid connectivity (on-prem to private IP).
Security	Secures traffic from VNet to service, but the service endpoint remains publicly accessible (can be firewall-restricted).	Highest security; traffic stays within VNet/Microsoft backbone, no public exposure; excellent for data exfiltration prevention.
On-premises access	No direct access (via public IP)	Yes (via VPN/ExpressRoute to private IP)
IP Address	Public IP of service (traffic uses private VNet IP to reach it)	Private IP from your VNet
DNS	No change needed (resolves to public IP)	Requires Private DNS Zone
Cost	Free (no added cost)	Cost for endpoint & data processed

Microsoft

AZ-104

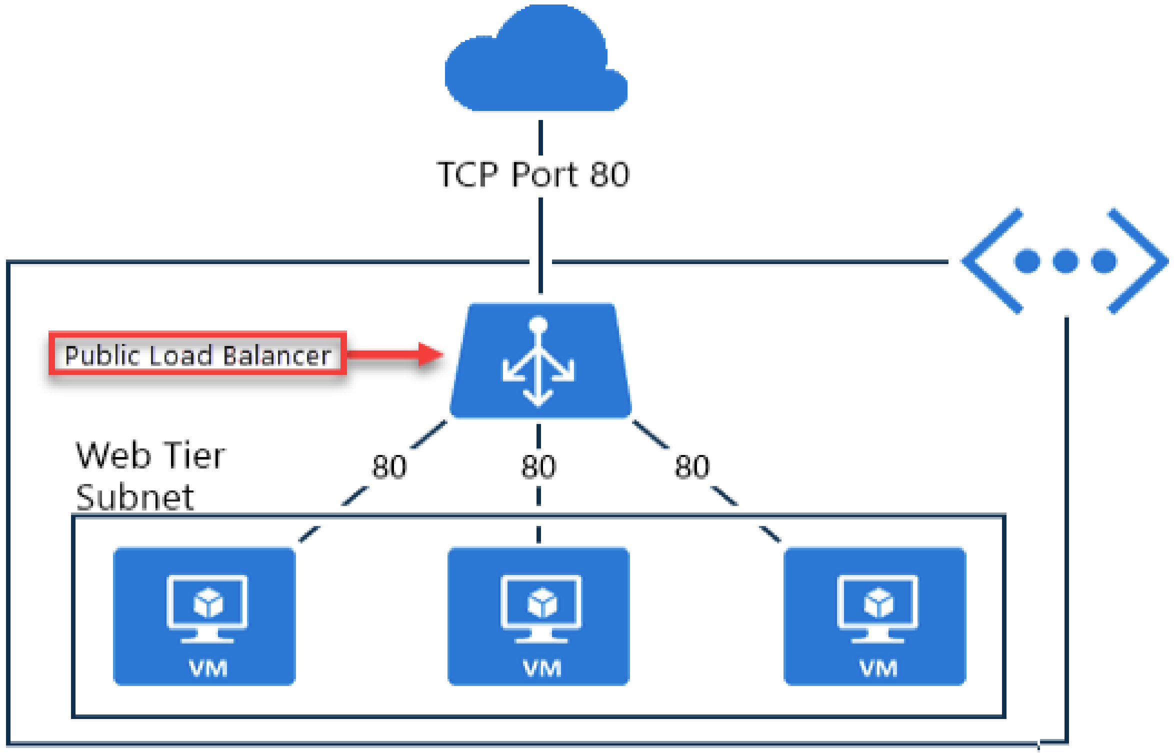
Administer Network Traffic

Choose a Load Balancer Solution

Feature	Application Gateway	Front Door	Load Balancer	Traffic Manager
Usage	Optimize delivery from application server farms while increasing application security with web application firewall.	Scalable, security-enhanced delivery point for global, micro service-based web applications.	Balance inbound and outbound connections and requests to your applications or server endpoints.	Distribute traffic to services across global Azure regions, while providing high availability and responsiveness.
Protocols	HTTP, HTTPS, HTTP2	HTTP, HTTPS, HTTP2	TCP, UDP	Any
Private (regional)	Yes		Yes	
Global		Yes		Yes
Env	Azure, non-Azure cloud, on premises	Azure, non-Azure cloud, on premises	Azure	Azure, non-Azure cloud, on premises
Security	WAF	WAF, NSG	NSG	

Implement a Public Load Balancer

- Maps public IP addresses and port number of incoming traffic to the VM's private IP address and port number, and vice versa
- Apply load balancing rules to distribute traffic across VMs or services

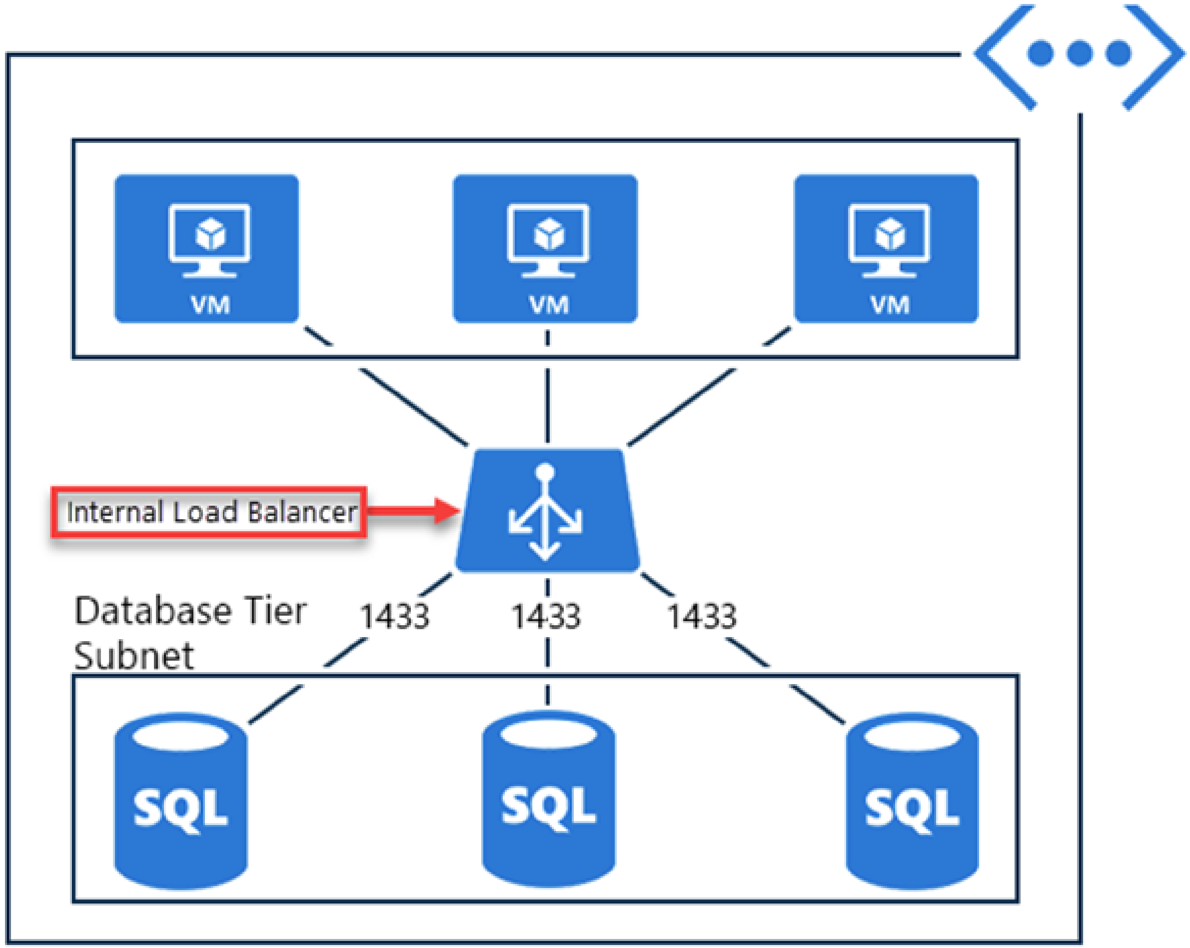


Implement an Internal Load Balancer

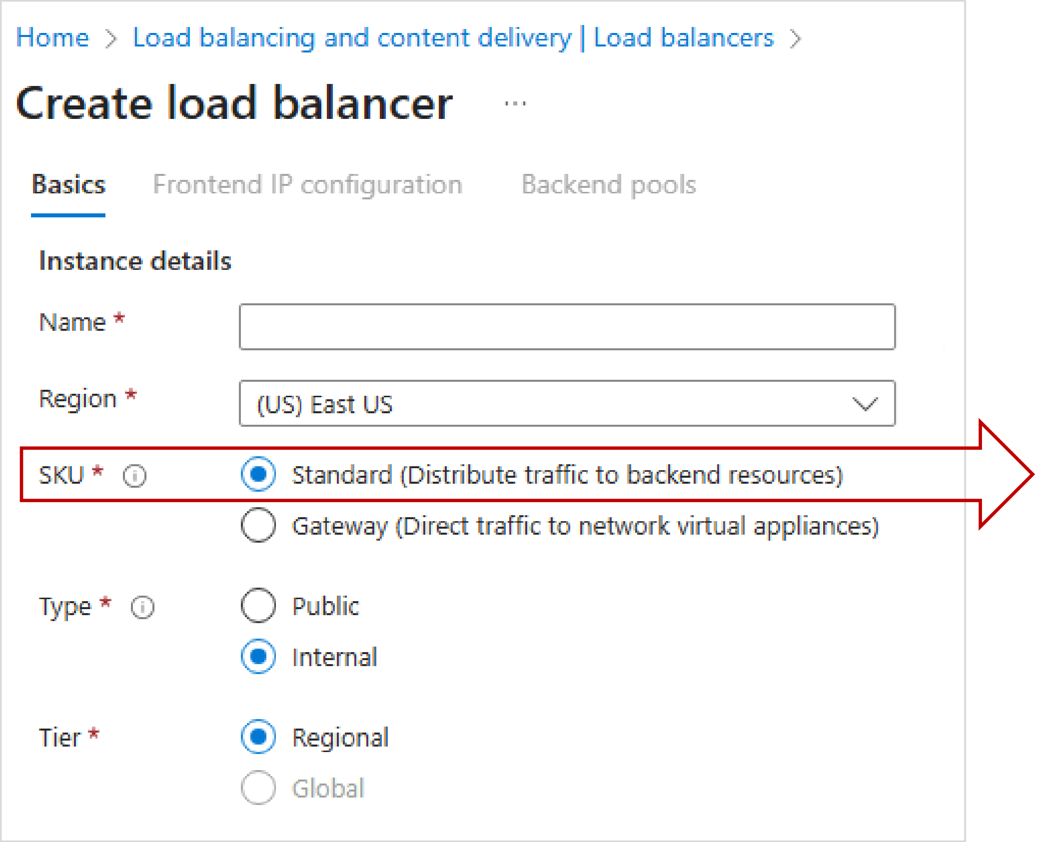
Directs traffic only to resources inside a virtual network or that use a VPN to access Azure infrastructure

Frontend IP addresses and virtual networks are never directly exposed to an internet endpoint

Enables load balancing within a virtual network, for cross-premises virtual networks, for multi-tier applications, and for line-of-business applications

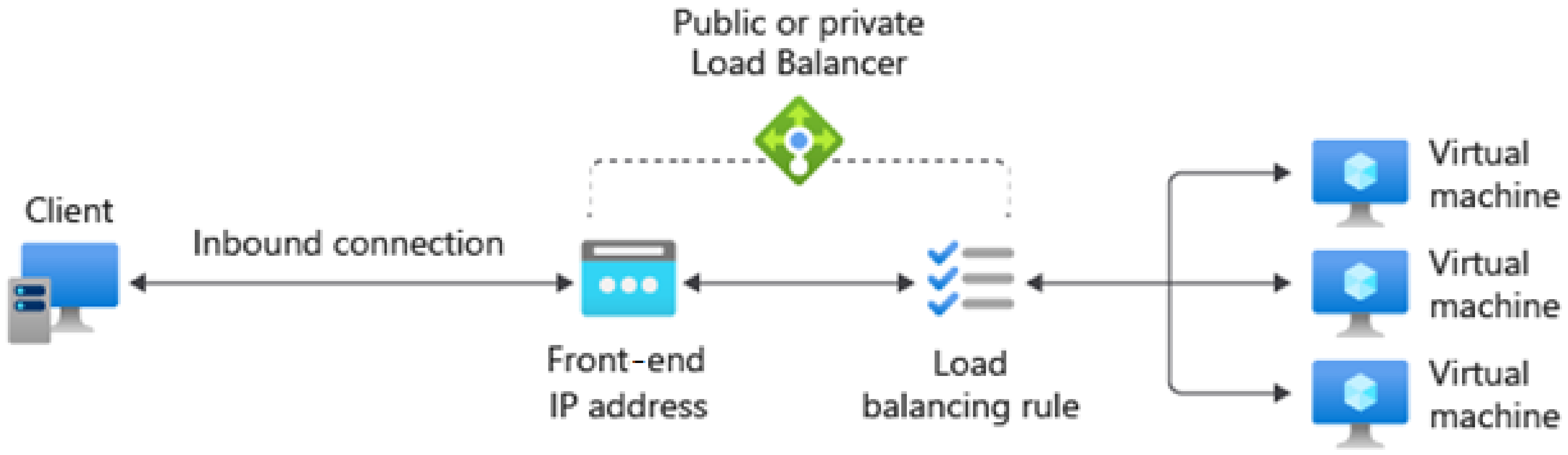


Determine Load Balancer SKU



Feature	Standard Load Balancer
Scenario	High performance, ultra-low latency, and high resiliency
Type	Internal or public
Backend type	IP based or NIC based
Multiple frontends	Inbound and outbound
Health probes	TCP, HTTP, HTTPS
SLA	99.99%

Create load balancer rules

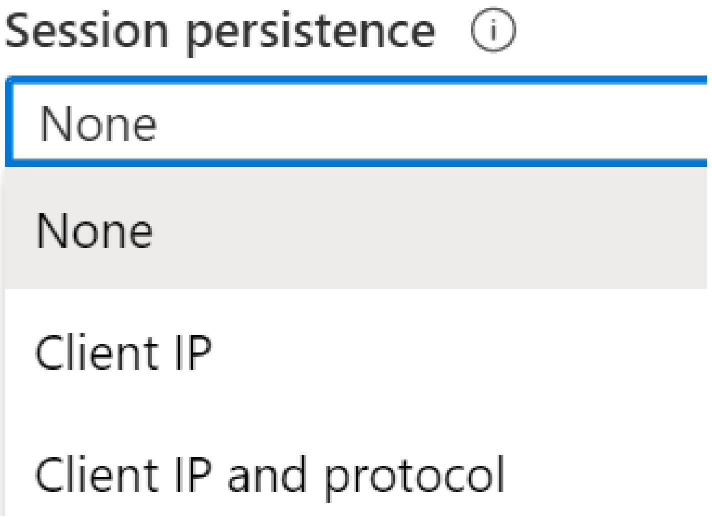


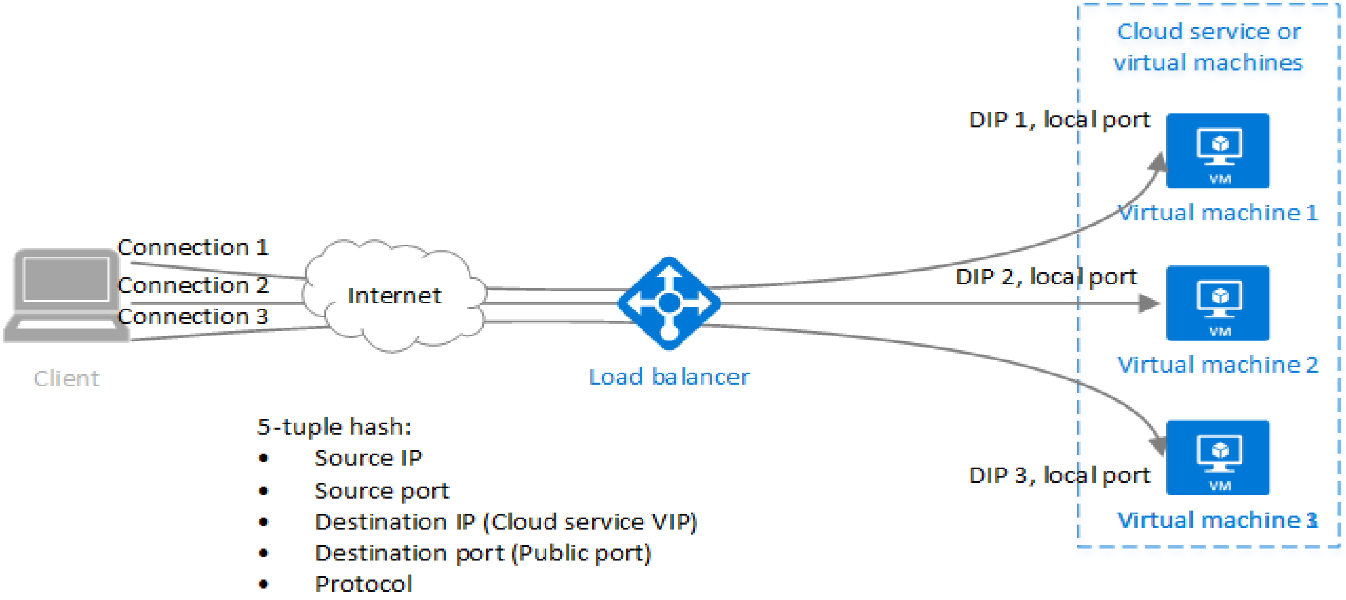
Maps a frontend IP and port combination to a set of backend pool and port combination

Rules can be combined with NAT rules

A NAT rule is explicitly attached to a VM (or network interface) to complete the path to the target

Configure Session Persistence (optional)





Session persistence specifies how client traffic is handled

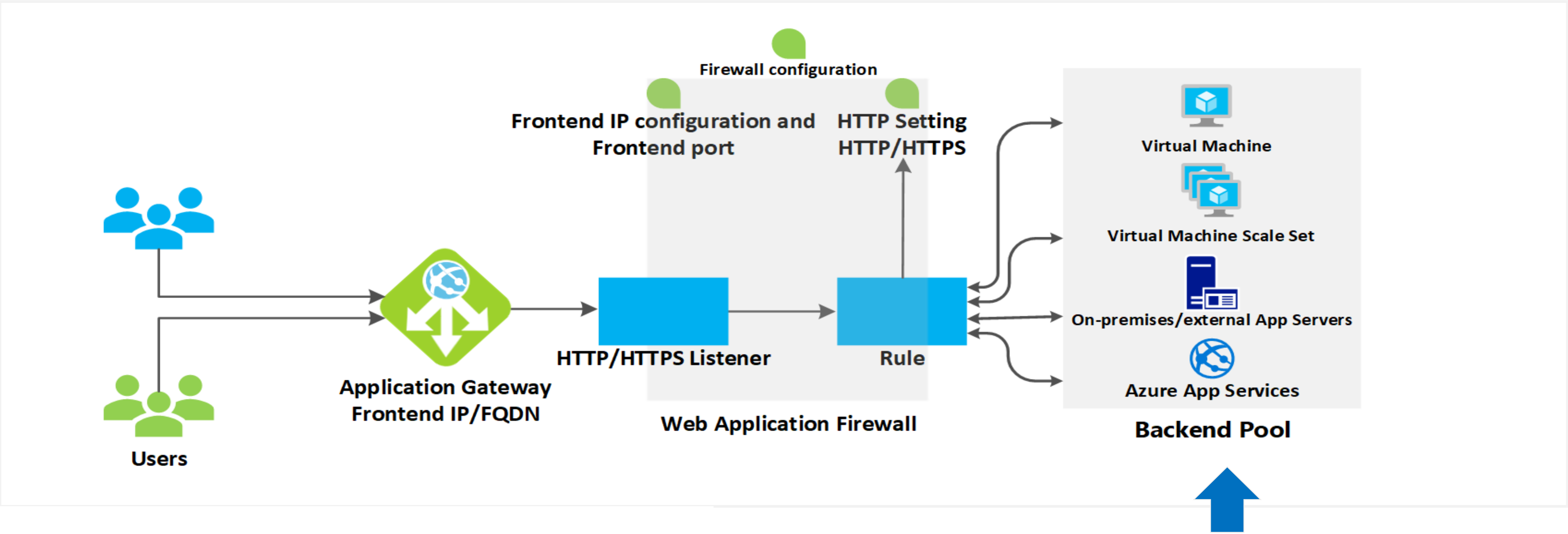
None (default) requests can be handled by any virtual machine

Client IP requests will be handled by the same virtual machine

Client IP and protocol specifies that successive requests from the same address and protocol will be handled by the same virtual machine

Introduction to Azure Application Gateway

Implement Application Gateway

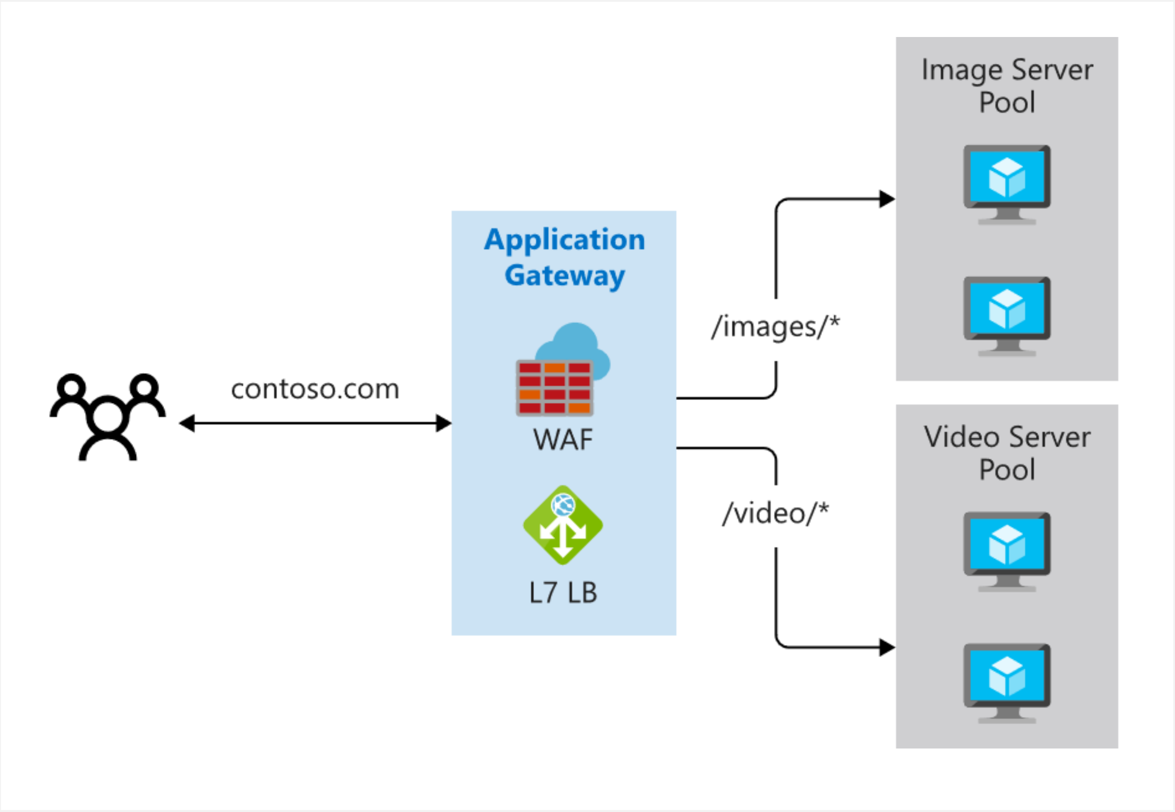


- Manages web app requests and routes traffic to a backend pool based on the URL

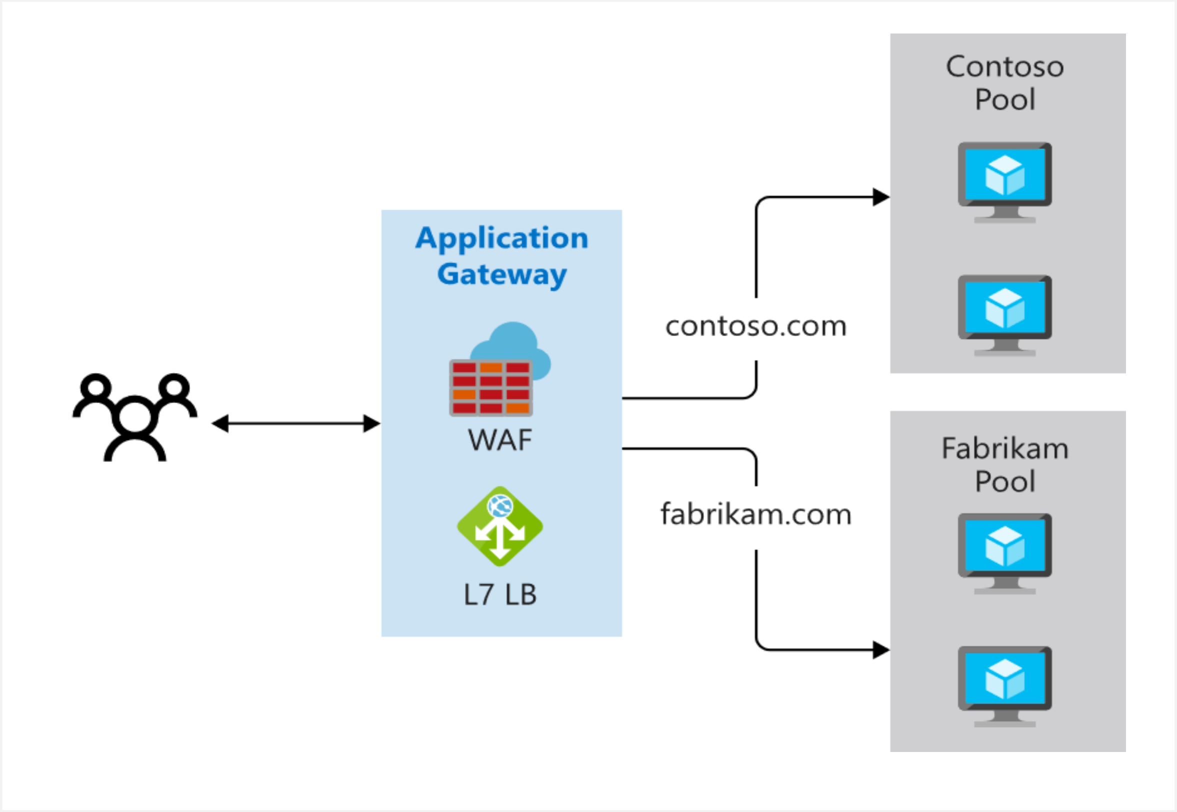
Azure virtual machines, Azure VM scale sets, Azure App Service, and even on-premises servers

Determine Application Gateway Routing

Path-based routing

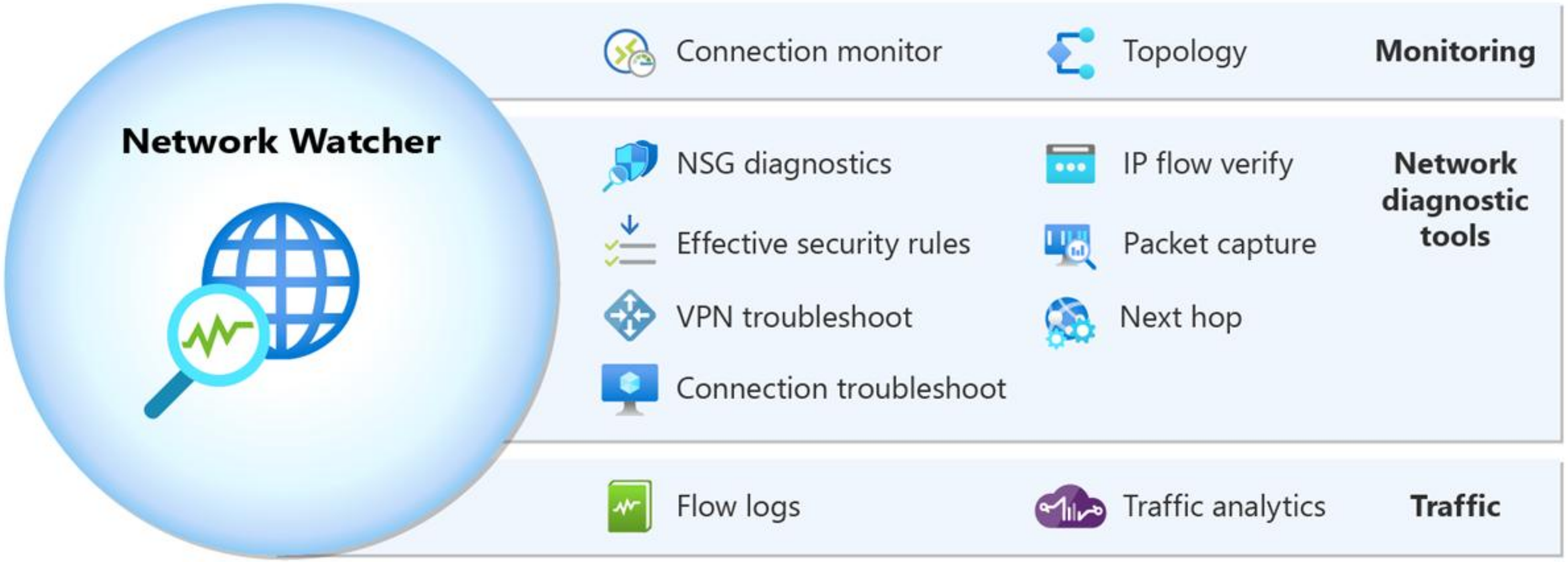


Multiple-site routing



Introduction to Network Watcher

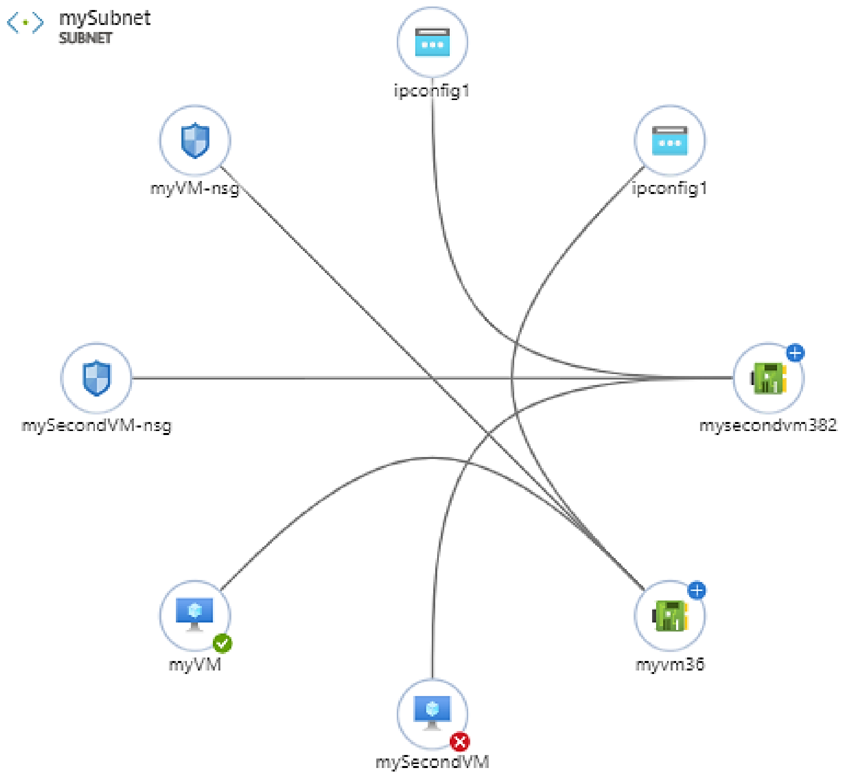
What is Network Watcher?



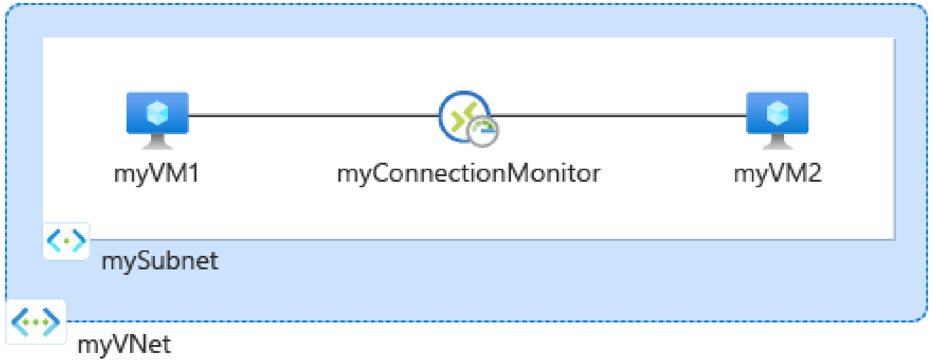
- Azure Network Watcher is a suite of tools
- Includes monitoring, network diagnostics, and traffic information
- Automatically enabled for every subscription.
- Can help repair the network health of IaaS products like VMs, VNets, application gateways, load balancers, etc.
- Network Watcher isn't designed or intended for PaaS monitoring or Web analytics.

Review Monitoring Tools

Topology provides a visual representation of your networking configuration

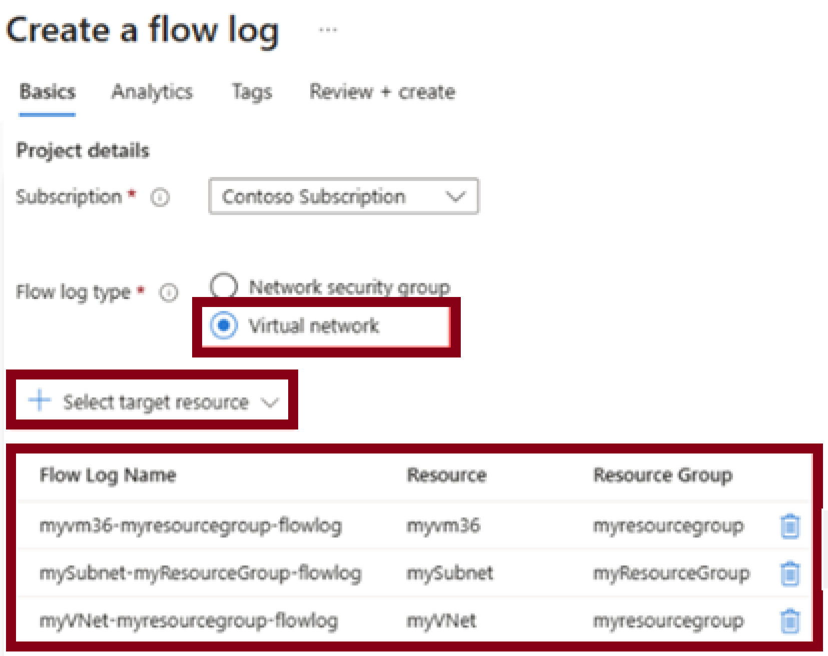


Connection Monitor provides end-to-end connection monitoring for Azure and hybrid endpoints



Review Traffic Tools

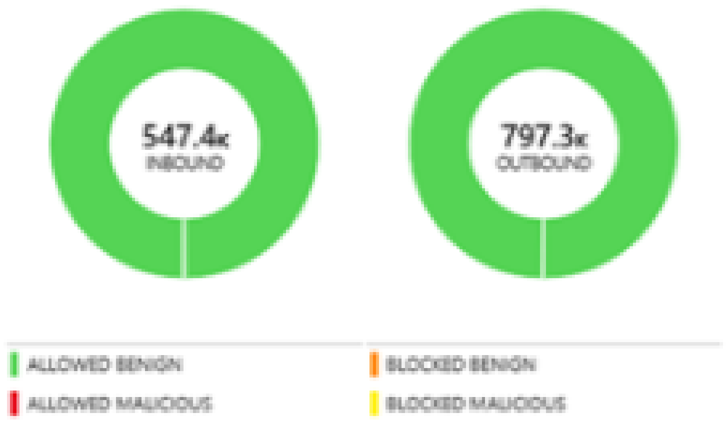
Flow Logs shows information on ingress and egress of IP traffic through a NSG or VNet



Raw Logs

Traffic analytics provides visualizations of flow logs data

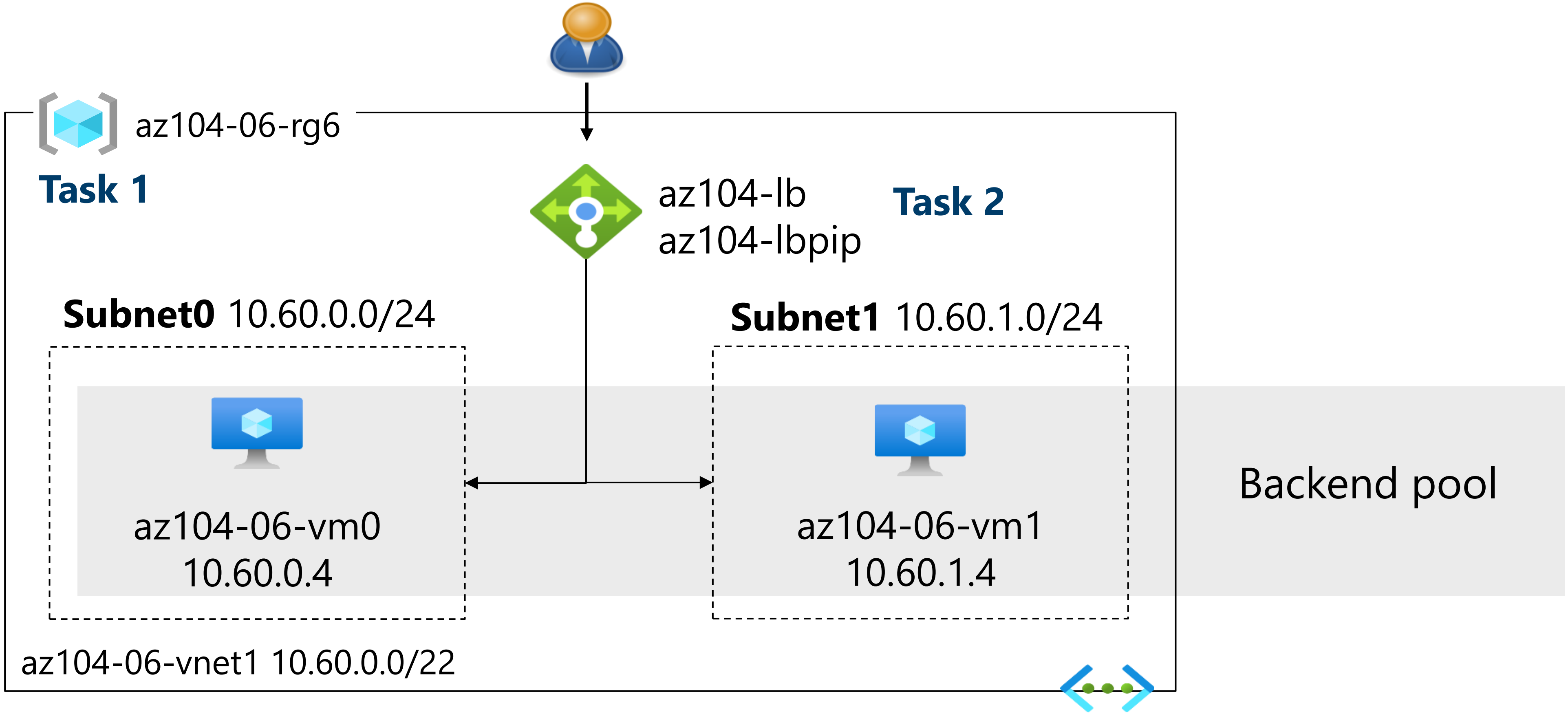
- Aggregate NSG and VNet flow log records
- Enhance the records with geography, security, and topology information
- Provide analytics – traffic hotspots, traffic distribution …



Labs

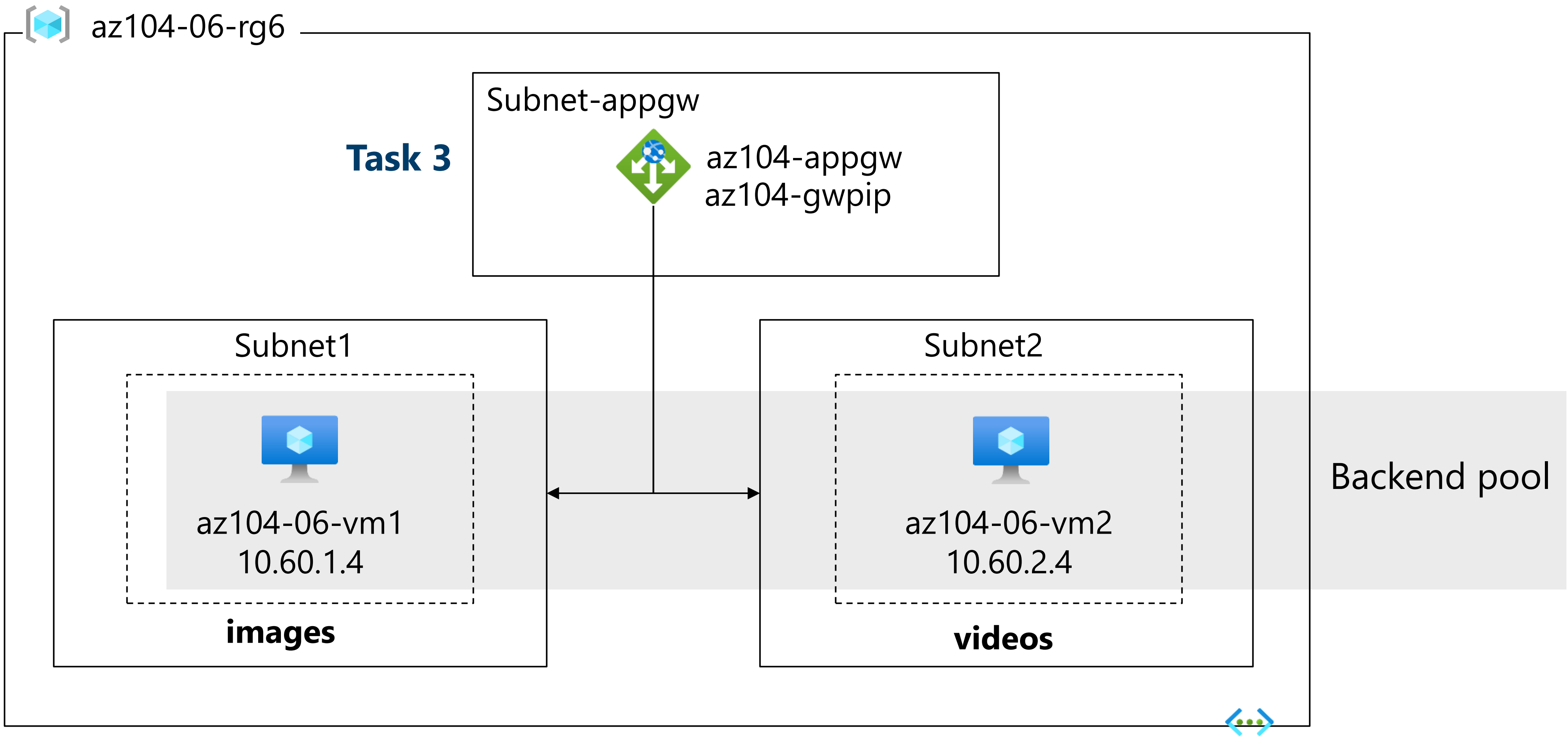
Lab 06 – Implement Traffic Management

Lab 06 – Architecture Diagram (Load Balancer)



Next slide for an application gateway

Lab 06 – Architecture Diagram (Application Gateway)



End of presentation